

AGUWA, CHINEDU

Kubwa, Abuja, FCT, Nigeria

Email: chineduaguwaofficial@gmail.com || Tel: +2348105471046 || LinkedIn: [\(Link\)](#) || GitHub: [\(Link\)](#) || Website: [\(Link\)](#)

EDUCATION

Federal University of Technology, Minna, Nigeria.

Oct 2018 - Feb 2025

Bachelor of Engineering (B.Eng.) in Civil Engineering

CGPA: 4.79/5.0 (3.89/4.0) | First Class Honours

WORK EXPERIENCE

Senior AI & Sustainability Platform Engineer

Apr 2026 – Present

EDAT Climate Data and Analytics — Houston, Texas (Remote)

- Promoted to lead development of new sustainability features on EDAT's platform, including reporting, benchmarking, and ESG data management modules, with ownership of architecture decisions, deployment, and production readiness for assigned platform areas.
- Built AI-powered forecasting and anomaly detection tools for sustainability insights across energy, waste, and emissions data, and designed APIs and data structures integrating finance, metering, supply chain, and audit information into a single platform.
- Supported customer demos and technical discussions during implementation, translating sustainability and customer requirements directly into engineering work.

Environmental Sustainability Software Engineer (Part-time)

Apr 2025 – Apr 2026

EDAT Climate Data and Analytics — Houston, Texas (Remote)

- Led backend architecture for a production-grade carbon accounting platform in Django, modelling Scope 1, 2, and 3 emissions across stationary/mobile combustion, fugitive emissions, purchased electricity, renewables, EPA WARM v16 waste accounting and water with 10+ activity-type ORM models feeding a centralised emission record system storing versioned GWP outputs as CO₂, CH₄, N₂O, and CO₂e.
- Built asynchronous processing pipelines with Django and Celery for large-scale emissions batch computation, GWP conversion, scheduled reporting, and AI-driven emissions classification and validation integrated directly into the ingestion pipeline.
- Collaborated on the Next.js/ShadCN frontend for Scope-level emissions dashboards and trend visualisation.

Full Stack Developer (Part-time)

Dec 2024 – Mar 2025

Eagle AI Labs — Dubai, UAE (Remote)

- Designed and implemented scalable REST APIs in NestJS and PostgreSQL for a crypto token platform, engineering tokenomics logic including rewards distribution, token vesting schedules, and multi-currency wallet management.
- Integrated RAG-based AI features for intelligent user assistance alongside multiple third-party crypto market APIs (price feeds, portfolio tracking) into a unified analytics dashboard, delivering the full backend-to-dashboard feature set within a 3-month remote contract.

Mobile Developer

Feb 2024 – Nov 2024

SureData Consulting Ltd — Tilbury, UK (Remote)

- Architected and shipped ChapelMate, a production-grade church management system for iOS and Android using React Native and Expo, published on both the Google Play Store and Apple App Store, with a Node.js/Express/PostgreSQL backend on Ubuntu implementing JWT authentication, RESTful API architecture, and real-time data synchronisation.
- Built ChapelMate's contribution tracking, PayPal payment integration, and push notification systems, giving church administrators membership and finance management in a single platform.
- Delivered feature enhancements to CallUp247 (Flutter, service-provider messaging and real-time chat) and contributed mobile features and third-party API integrations to Kint, a Flutter-based health management app handling patient data workflows.

Mobile & Full Stack Developer Intern

Nov 2023 – Jan 2024

Exinn Digital Technology — Maputo, Mozambique (Remote)

- Developed a full-stack medical web application supporting nursing mothers, using TypeScript, React, Node.js, NestJS, Supabase, and MongoDB, covering user authentication, clinical content delivery, and personalised health guidance based on user health data.
- Built a React Native patient-facing mobile app with an AI-powered chatbot for doctor-patient interaction and a Gemini-based clinical note generator that auto-drafts documentation from patient interactions.

Software Engineering Trainer *(Part-time)*

Jun 2023 – Oct 2023

A-Z New Age Tutor — Nigeria (On-site)

- Designed and delivered a structured practical training curriculum for 5 undergraduate students across a 4-month programme, covering Python, JavaScript, React, Node.js, and introductory machine learning applications.
- Built hands-on, project-based exercises translating core programming concepts into working code, with each student completing a functional end-to-end project by programme completion.

Software Developer Intern

Aug 2021 – Jun 2023

Scholarly Pathways — Nigeria (Remote)

- Designed, developed, and deployed a WordPress-based scholarship news platform for undergraduate students across Nigeria, implementing SEO optimisation and search indexing via Google Search Console, and integrated Google AdSense monetisation to generate revenue alongside the content mission.
- Maintained and iterated on the platform over a 22-month engagement, managing content architecture, plugin configuration, and performance optimisation.

SKILLS

Languages: Python · C++ · JavaScript · TypeScript · Dart · SQL · Go

Frontend: React · Next.js · Tailwind CSS · ShadCN UI · Bootstrap · Sass

Backend: Django · Django REST Framework · Node.js · Express · NestJS · FastAPI · Celery

Mobile: React Native · Expo · Flutter

Databases & Storage: PostgreSQL · MongoDB · MySQL · Firebase · Supabase · Neon

AI & Machine Learning: TensorFlow · Scikit-Learn · Pandas · NumPy · OpenCV · MATLAB · Random Forest · CNN · RAG pipelines ·

Cloud, DevOps & Tools: Docker · AWS · Azure · Ubuntu Server · CI/CD · Git · GitHub

RECENT PERSONAL PROJECTS

BuildCore — Sustainable Construction Management Platform

Next.js · Django · React Native · PostgreSQL ·

- Architecting BuildCore, a full-stack construction management platform (Next.js, Django REST Framework, React Native, PostgreSQL) integrating Bills of Materials, material inventory tracking, wastage management, and embodied carbon accounting into a unified project reporting system, with backend APIs supporting real-time material consumption logging and project-level carbon calculations aligned with LCA principles.
- Building a React Native field app for on-site material tracking and syncing, alongside an AI-driven material recommendation engine that suggests lower-carbon substitutes based on performance, cost, and environmental constraints.

Pavement Distress Detection System

TensorFlow · Flutter · FastAPI · Google Maps SDK

- Designed and trained a CNN on 30,000 labelled pavement images to classify distress types (cracks, potholes) with confidence-scored predictions, exported to TensorFlow Lite and optimised via an OpenCV preprocessing pipeline (resizing, normalisation, channel formatting) for fast on-device inference on resource-constrained mobile hardware.
- Built a full-stack Flutter app integrating the model with real-time GPS tagging, detection logging, and Google Maps visualisation of distress locations, backed by a FastAPI service handling authentication, detection history, geospatial analytics, and an admin dashboard for network-wide distress review.

Concrete Mix Ratio Predictor — ML Ensemble Model

Python · Scikit-Learn · Random Forest · Pandas

- Trained a MultiOutput Random Forest ensemble (300 estimators, max depth 12) on 712 real-world concrete mix records to predict aggregate-to-cement ratios ($R^2 = 0.91/0.90$, MAE = 0.08/0.09), using engineered features (paste volume, fine fraction, binder efficiency) and 5-fold cross-validation (mean $R^2 = 0.89 \pm 0.03$)

benchmarked against Linear Regression, Decision Tree, and XGBoost, then serialised with Joblib for downstream integration.

For more projects and details, visit [\(Link\)](#).

HONOURS AND AWARDS

Top 100 Africa Future Leaders — Class of 2025

2025

2025 Top 100 Africa Future Leaders Programme Selected from a competitive continental pool as one of 100 outstanding young African leaders demonstrating exceptional innovation and professional promise.

Vice-Chancellor's List — 37th & 38th Founders' Day

2022 & 2023

Federal University of Technology, Minna Recognised by the Vice-Chancellor on two consecutive occasions for outstanding academic performance, awarded to the highest-achieving students across the university.

Petroleum Technology Development Fund (PTDF) In-Country Scholarship

2021 – 2024

Petroleum Technology Development Fund, Federal Government of Nigeria Competitive national scholarship awarded to high-achieving engineering students across Nigerian universities, covering four consecutive years of studies.

CERTIFICATIONS AND ONLINE COURSES

AI & Machine Learning

CS221: Artificial Intelligence, Stanford Online

Python Data Structures and Algorithms

Software Engineering & Architecture

Software Architecture, Engineering and Development

Mastering WordPress Development and Monetisation

Analytics & Growth

Google Ads & Google Analytics 4

Google Search Console Mastery